# FaceVitals 2027 | Remote PPG Research Notebook

Algorithms, failure modes, synthetic baselines and real-video study plan

19 October 2026 - Phase 0

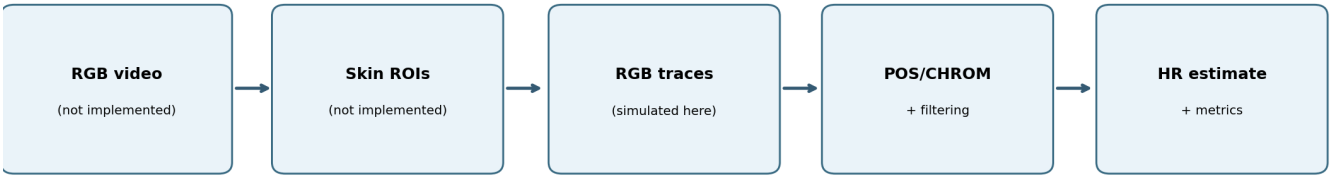

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**Prepared for IEEE Signal Processing Cup 2027.** This independent study kit is not an official IEEE publication.

**Evidence warning:** all experiment tables below describe generated RGB time-series only. No real video, measured lux, physiological SpO2/BP, or clinical measurements have been benchmarked by this kit.

Full rPPG pipeline and the actual boundary of this prototype




System diagram: implemented versus unimplemented pipeline steps

# 01 | A beginner-to-researcher guide to rPPG algorithms

**Goal:** understand how facial video can reveal heart rate, why competing algorithms behave differently, and why SpO2 and BP need more than a pulse peak.

# 1. What the camera actually measures

A conventional photoplethysmograph (PPG) uses a light source and detector to measure blood-related variation in transmitted or reflected light. Remote PPG (rPPG) observes tiny, time-varying color changes of skin under illumination, using a camera instead of contacting the skin. Each heartbeat changes blood volume near the surface; hemoglobin absorption and tissue scattering slightly change the reflected radiance. The camera converts it to R, G and B electrical/image signals.

At a single region of interest (ROI) such as a forehead or cheek, define for frame *t*:

`r[t] = mean red value over valid skin pixels`, `g[t] = mean green`, `b[t] = mean blue`.

A useful schematic observation model is:

`RGB[t] = static skin color + cardiac modulation[t] + illumination[t] + motion/reflection artifacts[t] + camera noise[t]`.

This is a **conceptual additive approximation**, not a full optical forward model. A more physical starting point is that pixel intensity integrates light-source spectrum × tissue reflectance × camera spectral sensitivity across wavelengths, with additional camera processing (exposure, white balance, compression, nonlinear gamma). For this reason ordinary JPEG/video RGB channels are **not** calibrated wavelengths.

**The difficult part:** pulse color variations may be far smaller than apparent variation caused by changing pose, light, exposure or motion. Extracting a periodic peak does not prove that it is the heart: fluorescent flicker, movement or cyclic camera corrections can have a similar frequency.

# 2. The complete processing chain

```
RGB VIDEO (timestamps, frames per second)
    |
    v
Find faces; designate + track target across frames
    |
    v
Find landmarks + skin-mask forehead and cheeks; exclude hair, eyes, mask
    |
    v
Compute robust RGB mean/median per region, per frame
    |
    v
Normalize slow color changes; combine color channels (GREEN/CHROM/POS/etc.)
    |
    v
Remove trend + band-pass pulse frequencies (example 0.7--3.0 Hz)
    |
    v
Get pulse spectral peak and/or interbeat spacing
    |
    v
HR = 60 x pulse_frequency_Hz; add quality / abstention gates
    |
    v
Compare to synchronized clinical reference; stratify failure by condition
```

The 0.7–3.0 Hz passband is approximately **42–180 bpm**. Frequency 1.25 Hz means `1.25 cycles/s × 60 s/min = 75 bpm`. **A passband is a design decision**: it can exclude bradycardia or tachycardia outside its bounds, and will alias or distort with insufficient frame rate. The official specification calls for **at least 30 fps** for the controlled baseline and includes a **10 fps** stress case (a 6 fps Nyquist floor covers 3 Hz in theory, but this does not guarantee acceptable accuracy).

### Why normalize by channel mean?

Suppose facial pixels of one ROI average `(R,G,B)=(110,84,70)` and a pulse varies green intensity by 0.4. The raw green variation is tiny relative to 84. A normalized green trace `g_norm[t] = g[t] / mean(g) - 1` measures *relative* deviations and is more comparable across cameras/subjects. However normalization does **not** remove every changing lighting artifact or create missing photometric information.

### What does a band-pass filter do?

An idealized sine wave at 1.2 Hz passes a 0.7–3.0 Hz bandpass. Slow brightness drift at 0.1 Hz is attenuated; high-frequency sensor noise is attenuated. But periodic head movement at 1.2 Hz passes **as well**, which is why color projection, tracking and robust spatial aggregation matter as much as filtering.

### How HR is estimated from the spectrum

An FFT or periodogram decomposes the extracted time series into candidate frequencies. Find the strongest reliable peak `f_peak` inside the allowed HR band; estimate `HR = 60 f_peak`. Prefer a Hann window, enough duration, reproducible frequency interpolation, and a mechanism to detect dominant motion harmonics. A 20-second observation contains approximately 0.05 Hz *independent* frequency spacing (`1/T`), equivalent to 3 bpm; zero-padding smooths the displayed periodogram but does not add new independent data. Longer windows stabilize spectral estimates but reduce responsiveness to changing HR.

## 3. GREEN – the easiest baseline

1. Detect visible skin and compute the mean green-channel time series.
2. Divide by its temporal mean; remove baseline/trend.
3. Band-pass and find the strongest HR-band spectral peak.

Why green? Under many visible illumination conditions, the green channel has a comparatively strong pulsatile contrast. **Advantages:** trivial computation, interpretable, little tuning. **Weakness:** any illumination/motion artifact in green can overwhelm the pulse. A pure green method is valuable as a control; do not assume it should win.

## 4. CHROM – exploit chrominance rather than brightness

Reference: de Haan & Jeanne (2013), *Robust pulse rate from chrominance-based rPPG*, DOI 10.1109/TBME.2013.2266196.

Given temporally normalized channels `r,g,b` within a short window, form:

`X = 3r − 2g`

`Y = 1.5r + g − 1.5b`

`alpha = std(X) / std(Y)`

`pulse_CHROM = X − alpha·Y`

The **idea** is to construct chrominance combinations where certain common illumination changes are suppressed, then scale them to balance nuisance contributions. Recalculate window statistics while moving along the video and overlap-add short window estimates. It is *not* simply selecting red and blue channels like a physical two-wavelength pulse oximeter.

**Advantages:** interpretable, no training labels, often handles motion/illumination variability better than GREEN. **Weakness:** depends on assumptions about light/color response and skin reflection; some illumination spectra break those assumptions. RGB values should be temporally well aligned, and short segments can produce unstable `std` ratios.

## 5. POS – orthogonal projection to skin-tone direction

Reference: Wang et al. (2017), *Algorithmic Principles of Remote PPG*, DOI 10.1109/TBME.2016.2609282.

Again let `(r,g,b)` be normalized RGB within a short window, and compute:

`S1 = g − b`

`S2 = g + b − 2r`

`alpha = std(S1) / std(S2)`

`pulse_POS = S1 + alpha·S2`.

The projection is designed to isolate pulsatile color variation away from the dominant skin-color/intensity direction. Standard POS operates on overlapping short windows (e.g., 1.6 seconds) followed by temporal combination/postprocessing. In words: **look at how green differs from blue and how combined green/blue differs from red; blend those views adaptively**.

**Advantages:** inexpensive, unsupervised, a strong primary real-video baseline. **Weakness:** vulnerable when the skin-color assumptions fail, reflectance signal is small, motion changes the color balance, or artifacts have a pulse-like spectral peak.

**The implementation supplied here** is a teaching-oriented 1.6-second Hann-overlap-add adaptation with 25%-length hop and a common Butterworth filter; it is not a bitwise reproduction of the POS or CHROM authors' exact reference implementations. Before publication, validate against official implementations in rPPG-Toolbox/pyVHR.

## 6. ICA and PCA – separating mixtures

**PCA (principal component analysis)** looks for directions in RGB space with maximum variance. It rotates the 3-channel observations into mutually orthogonal principal axes. Example: if head motion produces far more RGB variance than heartbeat, the **first** principal component might represent motion, not pulse. Selecting PCA-1 is therefore a deliberately weak comparator, not a state-of-the-art PCA implementation.

**ICA (independent component analysis)** assumes observations are mixtures of statistically independent underlying sources; it seeks combinations that are maximally non-Gaussian/independent. A component might resemble heartbeat, while another resembles illumination. In practice (i) independence assumptions are approximate, (ii) multiple artifacts can dominate, and (iii) one must select the *pulse-like* component without cheating by looking at reference HR.

We implemented **PCA-1**, not ICA. Real-data experiments should add a properly specified ICA baseline with component selection based only on allowed input/features.

## 7. PBV, LGI, SSR and neural approaches

- **PBV (blood-volume-pulse signature):** combines color channels using expected pulsatile relative amplitude; beneficial when its modeled optical signature is valid; errors arise under different lighting/ cameras.
- **LGI (local group invariance):** removes nuisance variation through an estimated low-dimensional color subspace and invariant representation; needs careful ROI preprocessing.
- **SSR (spatial subspace rotation):** analyzes how the local skin-pixel color subspace changes over time; can use spatial pixel variation that framewise RGB means discard.
- **DeepPhys:** learns a physiological signal using image appearance + frame differences and learned attention; train/test leakage and changing camera pipelines matter.
- **PhysNet:** spatiotemporal video model learns waveform directly from video clips; typically heavier and label-dependent.
- **TS-CAN / EfficientPhys:** architectures emphasizing computational efficiency for physiological measurement; potentially more suitable to edge deployment than large 3-D models, subject to measured device latency/power.
- **PhysFormer:** attention-based temporal and spatial modeling; can be performant but is not an automatic edge/fairness solution.

Neural methods need data-quality checks, matching input representation, licensing and **subject-disjoint, device-disjoint evaluations**. Comparing a pretrained model evaluated on its training subjects to a new unsupervised baseline is not scientifically valid.

## 8. What exactly is a “failure rate”?

First define a test *unit*: a 20-second clip, fixed-person HR label, synchronized reference. Then define a threshold, for example `failure = |estimated HR - reference HR| > 5 bpm` or missing/invalid estimate. Across `N` clips, `failure_rate = number_failed / N` (×100 for percent). Separately report **MAE**, **RMSE**, **Pearson r**, **Bland-Altman bias** and limits (`mean difference ±1.96 × sample SD`) and *number of predictions rejected*. Always retain results by condition and demographic group; reporting only the pooled mean can hide systematic harm. The official challenge lists MAE, RMSE, Pearson and Bland-Altman; our >5 bpm threshold is an additional engineering measure.

There is no scientifically transferable figure such as “POS has a 12% failure rate” without specifying dataset, population, pre-processing, video quality, HR range, thresholds and ground truth. A published result on one dataset cannot be silently used as another dataset’s benchmark.

## 9. Why SpO2 is a separate science problem

Classical pulse oximeters measure **red and near-infrared wavelengths** with known spectral responses, using a calibrated AC/DC intensity ratio. A conceptual ratio-of-ratios is:

`R = (AC_red/DC_red) / (AC_IR/DC_IR)`.

The clinical device maps `R` to SpO2 through a calibrated relationship, typically including empirical human calibration. **Ordinary RGB videos do not contain a dedicated infrared channel**, and RGB spectral responses/white balance change with device/illumination. One may attempt **calibrated RGB-ratio regression**, or a jointly learned feature network using actual clinical SpO2 labels, but the model must prove performance on genuinely varied oxygen saturations, skin tones and previously unseen cameras. Ordinary datasets of healthy subjects at 97–100% oxygenation may support a trivial near-constant predictor; such performance does not demonstrate hypoxemia detection.

Do not report SpO2 as clinically accurate without cross-device, low-saturation, agreement and calibration validation. Source: *RGB-Only rPPG for Contactless SpO2 Estimation: A Critical Review* (2026), https://www.mdpi.com/2079-9292/15/15/3477 .

## 10. Why BP is not simply HR plus a formula

BP varies with peripheral resistance, arterial compliance, cardiac output, posture, physiological state and other factors that **cannot be uniquely determined from HR**. One possible research pathway is to extract waveform-shape features (systolic upstroke duration, notch/shape proxies, width, variability), combine them with video-derived and validated covariates, and train systolic and diastolic regressors against properly synchronized **cuff/reference BP labels**. Assess individual calibration vs calibration-free methods separately. A model that learns a person's average pressure can appear strong without tracking BP changes, so use within-person change tests as well as subject-disjoint validation.

A single RGB-derived pulse trace is not a measured pulse-transit time (PTT); don't label it PTT unless two actual timing locations and a validated temporal reference are available. Clinical BP claims require dedicated validation. The American Heart Association cautions that cuffless BP estimates can be calibration-dependent and are not yet interchangeable with standard diagnosis/management measures: https://professional.heart.org/en/science-news/cuffless-devices-for-the-measurement-of-blood-pressure .

## 11. Worked mini-example: interpreting agreement

Suppose four synchronized clips have reference HR `[60, 80, 100, 120]` bpm, while a method gives `[61, 78, 106, 119]` bpm. Errors (prediction minus reference) are `[1, -2, 6, -1]` bpm. MAE is `(1+2+6+1)/4 = 2.5` bpm, despite **one failure** under the >5 bpm rule (25%). RMSE = `sqrt((1+4+36+1)/4) ≈ 3.24` bpm. These summarize different aspects: MAE typical absolute magnitude; RMSE sensitivity to outliers; failure rate risk of large errors. Report all three rather than just an attractive MAE.

## 12. First real-video task after this package

On authorized UBFC-rPPG, PURE, organizer, or AiPEX videos, implement: video timestamp loading, target-face detection, forehead/cheek ROI selection, skin-pixel mean extraction, GREEN/POS/CHROM, reference synchronization and a reproducible evaluation. Freeze that protocol **before** testing novel additions. The generated prototype does **not** yet execute this step.

# 02 | Algorithm selection, evidence and dataset access

**Research status as of 19 October 2026.** This matrix is a *methodological triage*, not a pooled meta-analysis and not a set of established per-algorithm failure rates.

## Method comparison: where each approach can shine

| Method | Principle | Strengths | Main potential failure mechanisms | Edge feasibility | Phase-1 role |
|---|---|---|---|---|---|
| GREEN | green ROI mean + temporal filtering | transparent; extremely fast; useful check of camera/ROI | pulse-like motion, flicker, specular changes | Excellent | must-have control |
| CHROM | normalized RGB chrominance projections with adaptive scaling | interpretable; attempts illumination/ motion suppression | illumination spectrum and camera RGB response mismatch; unstable short-window SD | Excellent | primary baseline |
| POS | standardized color projection orthogonal to skin direction | good low-compute candidate; unsupervised; easy per-ROI | tiny pulse amplitude; changing skin reflection; heavy motion | Excellent | **primary starting baseline** |
| ICA | independent source separation in RGB | flexible nuisance mixing | independence violated; output component selection may track artifacts | Good | second-stage baseline |
| PCA | largest RGB variance or selected PCs | easy dimension reduction | largest component can be motion rather than pulse | Excellent | diagnostic comparator, not PCA-1 as best method |
| PBV | estimated blood-volume color signature | physiology-inspired projection | model mismatch under varied light/ camera/skin | Excellent | ablation candidate |
| LGI / SSR | invariant/spatial color subspace methods | may exploit pixel structure and nuisance subspaces | ROI quality, illumination, insufficient skin pixels | Good | advanced classical |
| DeepPhys | CNN learned from appearance + changes | can learn spatial attention under representative training | bias, compression generalization, labels needed | Moderate | deep baseline |
| PhysNet | spatiotemporal learned waveform | captures nonlinear time/space cues | heavy deployment; high-data needs | Lower | GPU reference |
| TS-CAN / EfficientPhys | efficiency-oriented neural signal estimation | potentially useful on edge | inference cost still needs hardware testing; out-of-domain shift | Moderate/Good | later edge candidate |
| PhysFormer | temporal/spatial attention | expressive representation | compute/data cost, skin-tone & camera domain shift | Lower | aspirational higher-capacity reference |

**Recommendation:** Begin with **GREEN, POS and CHROM**, with consistent face/ROI preprocessing, same HR-band filter and fixed evaluation clips. After real-video ground truth is acquired, add ICA/PBV, patch-level aggregation and a model from rPPG-Toolbox. No method is an unconditional “best” across conditions; even the apparently best one in the synthetic CSV may lose on authentic videos.

The `src/benchmark.py` prototype implements GREEN, CHROM, POS and PCA-1 on *synthetic ROI RGB traces*. It does not implement any of the deep methods, ICA, PBV, LGI or SSR.

## Why we are not copying “failure rates” out of papers

A defensible method-to-method error comparison must use the **same actual clips, reference-signal synchronization, skin ROI, test splits, HR-estimator configuration and failure definition**. Different papers often use different datasets and metrics (MAE, RMSE, correlation, signal-to-noise ratio, fraction with errors >5 bpm). Even if paper A reports 2 bpm MAE and paper B reports 5 bpm MAE, that alone does not prove A’s algorithm is better.

### Literature-grounded benchmark references

- **Wang et al., 2017, POS mathematical foundations:** https://pubmed.ncbi.nlm.nih.gov/28113245/
- **de Haan & Jeanne, 2013, CHROM:** https://pubmed.ncbi.nlm.nih.gov/23744659/
- **pyVHR:** cross-dataset workflows for GREEN, ICA, PCA, CHROM, POS, SSR, LGI, PBV, OMIT; https://github.com/phuselab/pyVHR
- **rPPG-Toolbox:** public reference implementations and training/evaluation workflows; https://github.com/ubicomplab/rPPG-Toolbox
- **AiPEX-Lab diversity/bias study:** real videos from India and Sierra Leone, published method-comparison CSV, source code and methodology; https://github.com/AiPEX-Lab/rppg_biases ; associated paper https://doi.org/10.1038/s41746-021-00462-z . The paper reports variability and bias under its *own protocol*; do not mislabel this as a generic per-method failure percentage.
- **RGB-only SpO2 critical review (2026):** https://www.mdpi.com/2079-9292/15/15/3477
- **AHA cuffless BP scientific statement:** https://professional.heart.org/en/science-news/cuffless-devices-for-the-measurement-of-blood-pressure

## Dataset ranking and what is actually measured

| Dataset/source | Best use | Critical caution | Access |
|---|---|---|---|
| **Organizer’s 2027 clinical dataset** | direct challenge alignment, three-vital validation if labels match | requires copyright/legal agreement; obtain acquisition metadata | email Dr. Vijay Jeyakumar (copy competition list) |
| **AiPEX-Lab rppg_biases** | subgroup and cross-condition *HR* comparisons; multiple facial ROI clips | publicly provided means/ videos should be checked for timing, actual pixel availability and ground-truth format | GitHub repository |
| **UBFC-rPPG** | foundational real-video HR + PPG test | Dataset 1/2 labels differ; not a definitive BP or hypoxemia benchmark | public dataset page |
| **PURE** | movement and controlled recording HR baseline | verify availability/usage terms and exact reference-signal format | request authors / public routes where permitted |
| **VIPL-HR** | broader devices/lighting and pulse labels; investigate SpO2 availability per session | changing hardware & capture conditions require subject/ device splits | published dataset site |
| **COHFACE** | illumination and real PPG/ respiration comparison | limited cohort; older camera | dataset owners/site |

| Dataset/source | Best use | Critical caution | Access |
|---|---|---|---|
| **BP4D+** | extensive subject-diverse facial video and physiological/ cuff pressure-wave modalities | not an interchangeable clinical SpO2/ECG/PPG ground truth; must inspect labels and synchronization | faculty-led license request |
| **BP4D++** | more age diversity including participants through 70; physiology and video | large, access-restricted; ascertain which BP labels and reference techniques actually provided | inquiry to Dr. Lijun Yin |
| **BU-3DFE** | static 3D expression/face geometry only | **not suitable as a core HR/ rPPG dataset** because static expression models lack synchronized pulse video/ reference | not a priority |
| **Camera Vitals website** | discovery index of datasets, tools and papers | **not itself a dataset** | https://cameravitals.github.io/ datasets.html |

**Important correction to dataset search:** The Binghamton page is titled "3D Facial Expression Database" but covers *several* different datasets. Request **BP4D+ or BP4D++**, not BU-3DFE, for relevant physiology; the site says students cannot individually receive BP4D+ and asks a **faculty supervisor** to request data and sign institutional agreements. The site also describes a substantial external-drive transfer for BP4D+. **Ask whether a smaller authorized subset is available** before planning transfer.

## Competition specification notes

- Official source: https://signalprocessingsociety.org/sites/default/files/2026-10/2027-SP-Cup-Official-Document-FaceVitals.pdf
- Round 1: HR under controlled and uncontrolled conditions; at least two robustness constraints addressed and justified.
- Round 2: HR + SpO2 with pulse oximeter reference.
- Final phase: HR + SpO2 + systolic/diastolic BP, clinically benchmarked and live demonstration.
- Stress requirements include ≤50 lux; sub-HD down to QVGA; ≤500 kbps and ≤10 fps; ages 60–70+; partial lower-face masking; multi-person selection; Fitzpatrick/skin-condition generalization.
- Official output convention includes `.mat` payloads, paired RGB video, reference and processed CSV; metadata include light, age, skin type, camera FPS, distance, constraints.

**Documentation inconsistencies to clarify with organizers:** The technical document describes *three rounds* but also lists directory `/Round4_FullSystem_Clinical/`; the general uploaded Terms & Conditions lists **six** equally weighted final judging criteria, while an SPS web page currently displays **five**. Follow the challenge-specific instructions and request written clarification rather than guessing submission folder names or scoring weights.

# 07 | First results: synthetic RGB-trace benchmark

**WARNING: illustrative simulations only. None of these percentages measure clinical or real-video performance.**

Generated 210 artificial traces, evaluated by 4 algorithms = **840 estimates**. Each scenario has 35 independent artificial traces with randomized HR (52–138 bpm). One output per method per artificial trace. Reproducible random seed: 2027.

## Metrics table

The failure definition is `abs(est-true)>5 bpm OR missing`; MAE/RMSE are in bpm. Correlation and limits of agreement are also recorded in the CSV.

| Scenario | Method | MAE (bpm) | RMSE (bpm) | Failure >5 bpm / missing | n |
|---|---|---|---|---|---|
| controlled | PCA | 0.04 | 0.06 | 0.0% | 35 |
| controlled | GREEN | 0.04 | 0.05 | 0.0% | 35 |
| controlled | POS | 0.07 | 0.08 | 0.0% | 35 |
| controlled | CHROM | 0.07 | 0.08 | 0.0% | 35 |
| low_light_proxy | GREEN | 0.37 | 0.46 | 0.0% | 35 |
| low_light_proxy | PCA | 2.15 | 7.59 | 5.7% | 35 |
| low_light_proxy | CHROM | 17.79 | 28.16 | 42.9% | 35 |
| low_light_proxy | POS | 19.14 | 29.08 | 48.6% | 35 |
| motion_artifact | CHROM | 0.06 | 0.10 | 0.0% | 35 |
| motion_artifact | POS | 0.08 | 0.15 | 0.0% | 35 |
| motion_artifact | GREEN | 20.68 | 27.94 | 82.9% | 35 |
| motion_artifact | PCA | 20.69 | 27.94 | 82.9% | 35 |
| color_illumination | PCA | 34.46 | 41.14 | 91.4% | 35 |
| color_illumination | CHROM | 34.46 | 41.15 | 91.4% | 35 |
| color_illumination | POS | 34.46 | 41.14 | 91.4% | 35 |
| color_illumination | GREEN | 34.46 | 41.14 | 91.4% | 35 |
| 10fps_trace | GREEN | 0.03 | 0.04 | 0.0% | 35 |
| 10fps_trace | PCA | 0.03 | 0.04 | 0.0% | 35 |
| 10fps_trace | POS | 0.07 | 0.09 | 0.0% | 35 |
| 10fps_trace | CHROM | 0.07 | 0.09 | 0.0% | 35 |
| weak_pulse_proxy | PCA | 0.13 | 0.19 | 0.0% | 35 |
| weak_pulse_proxy | GREEN | 0.14 | 0.20 | 0.0% | 35 |
| weak_pulse_proxy | POS | 1.81 | 5.67 | 8.6% | 35 |
| weak_pulse_proxy | CHROM | 3.51 | 9.53 | 14.3% | 35 |

## Why these outcomes appear

**Controlled and 10 fps trace:** Strong artificial pulse with little in-band interference is easy for most spectral estimators. Downsampling a perfectly aligned pre-aggregated trace is *not* the same as feeding a poor 10 fps smartphone video into a face detector.

**Motion-artifact scenario:** The injected shared-intensity sinusoid is designed to be suppressible by normalized color projections. This benefits POS/CHROM under this *particular* artifact construction and exposes GREEN/PCA-1 to nuisance dominance. It does not prove real movement immunity.

**Low-light proxy:** The chosen absolute read-noise floor makes two-window color projections relatively unstable while the relatively stronger green pulse can remain visible. This is a *model effect*, not evidence that GREEN generally wins at 50 lux. Real cameras alter their exposure, gain, compression and temporal sampling.

**Color illumination:** Periodic chromatic interference falls inside the HR passband and defeats all tested methods frequently. Peak power may be large even when the predicted HR is wrong: spectral peak-to-median PSD is **not calibrated confidence**.

**Weak-pulse proxy:** This tests a deliberately smaller optical modulation, not pigmentation, Fitzpatrick classification, melanin absorption or any demographic attribute. It cannot support skin-tone fairness claims.

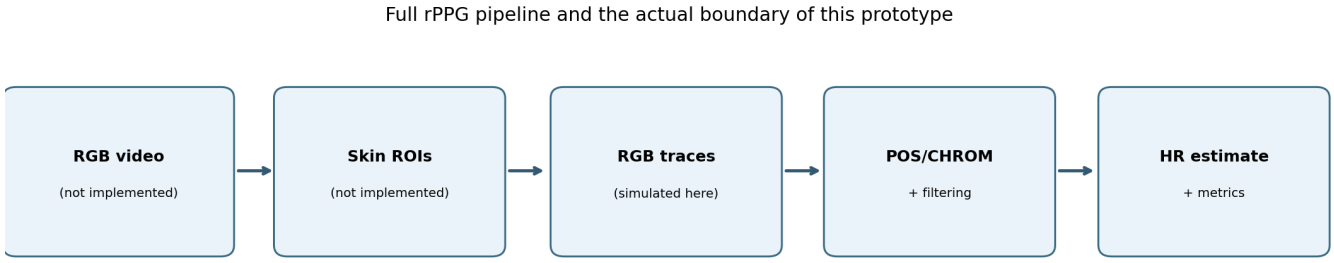
## Technical limitations

- Simplified, short-window CHROM/POS overlap-add variants; not exact original authors' implementation.
- No facial video decoding, face detection, ROI segmentation, real frame-drop motion, physical luminance or bitrate modeling.
- The model uses one artificial pulse shape across conditions; deliberately simple and **not calibrated** to human skin optics or real camera response.
- PCA-1 is not a full pulse-aware PCA selector; algorithms were not tuned or learned using human data.
- Per-condition sample size 35; no inferential real-population uncertainty analysis.
- A changing heart rate/arrhythmia is not simulated. No physiological SpO2/BP predictions were made.

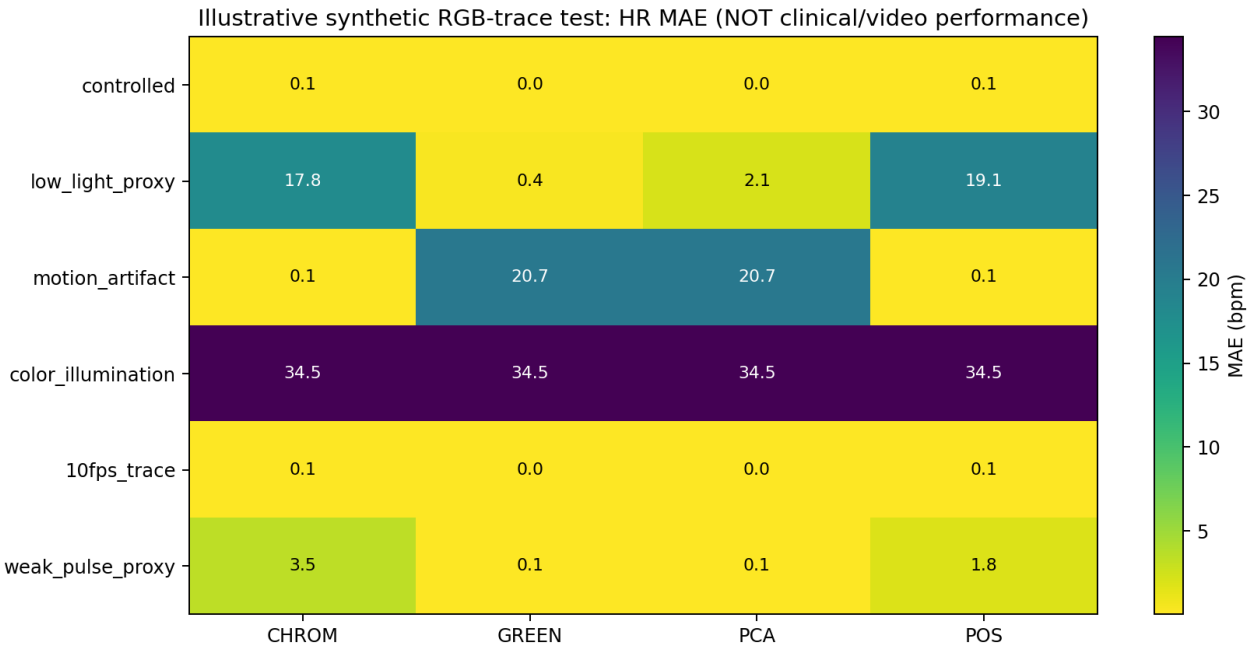
## Full reproducibility

`src/benchmark.py`, `tests/test_benchmark.py`, `results/manifest.json`, `results/synthetic_predictions.csv` and `results/synthetic_summary.csv` contain the complete generative assumptions, code, individual predictions and aggregations. Timing columns reflect only processing an RGB trace, **not end-to-end camera FPS**.

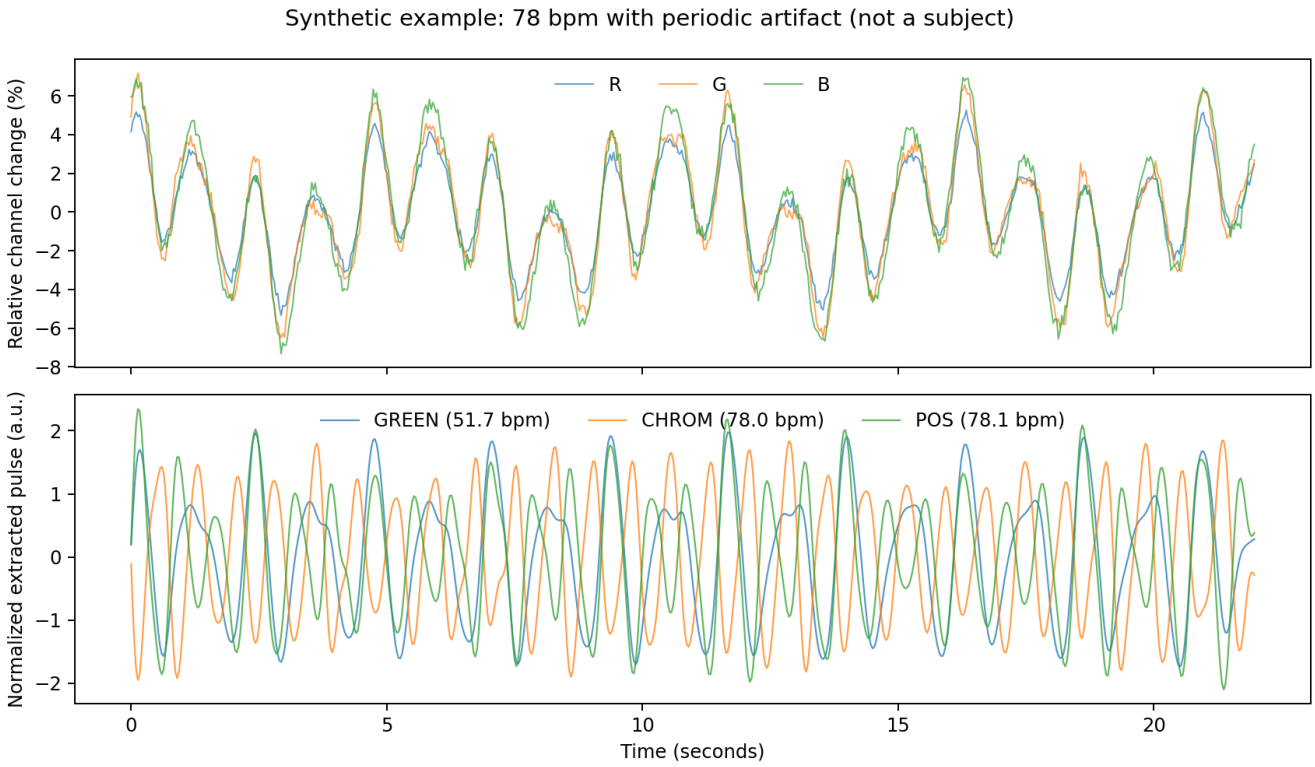
# Figures




Pipeline, highlighting absent image-processing stages




Synthetic MAE heatmap




Artificial RGB traces and extracted channels

Next required evidence: real RGB video with synchronized ground truth, well-documented consent/ licensing and participant-held-out evaluation.

# 03 | Reproducible benchmark protocol for real videos

## Study questions

- Q1: On real videos, which of GREEN, POS, CHROM, ICA and PBV best estimates **HR** under controlled conditions?
- Q2: Which degrades least under C1 low light, C4 low resolution, C5 reduced bandwidth/FPS, and C7 occlusion?
- Q3: Does **worst-group** HR error increase across objectively documented pigmentation strata, age bands or camera conditions?
- Q4: Is an uncertainty/quality gate capable of preventing confident wrong HR estimates without rejecting every hard sample?
- Q5 (later): Can SpO2 / systolic+diastolic BP be estimated with defensible calibration and subject-independent testing?

# 1. Before any model is trained or tuned

Create a reproducible sample manifest containing: `dataset_id, subject_id, clip_id, video_path, camera_model, frame_rate, timestamps, reference_type, reference_file, skin_tone_method, age_band, condition, measured_lux, video_bitrate, resolution, consent_license_status`. Unknown fields should be null, not invented. Securely store identifiable face videos and share only data permitted by licenses/consent.

Define **participant-disjoint splits**. A person's recordings MUST NOT appear across train/validation/test, and neighboring overlapping windows of the same person must not leak across splits. If testing camera generalization, hold out **entire cameras**. For robustness claims, evaluate the shift from baseline on matched subjects when possible, and report condition sample sizes. Calibrate thresholds/models only on training/validation, never the final test.

# 2. Input and face/ROI preprocessing

1. Decode exact frame timestamps, keep dropped-frame indicators; do not assume perfect constant FPS.
2. Detect *all* faces; select target ID and track across time (C3), measure identity-switch rate.
3. Extract forehead, left cheek, right cheek masks from landmarks; exclude hair, sclera, specular reflections and occlusions.
4. Average robust RGB over each skin mask per time step; store 3 traces and mean skin-pixel coverage.
5. Compute method output per ROI and optionally aggregate using **a score evaluated without reference HR** (e.g., spectral peak stability, patch consistency). Report any abstentions.
6. Process windows of 20–30 s to establish an initial baseline. Vary duration as a controlled ablation (e.g., 10/20/30 s).
7. Pass the *same* HR estimator and passband (for example 0.7–3 Hz) to every method. Publish latency after warm-up and memory per device.

# 3. Baseline & stress tests

| Test | Grounded competition target | Real experiment | Current synthetic prototype |
|---|---|---|---|
| Controlled | ≥300 lux, ≥30 fps, HD, 40–60 cm, frontal | measured setup + reference ECG/PPG | simulates clean RGB traces **only** |
| C1 low light | ≤50 lux | measure lux at face plane and capture real camera noise/exposure settings | low `relative_light_gain`; **not lux-calibrated** |
| C2 skin tone | multi-tone and dermatological variation | obtain consented Fitzpatrick/ITA metadata and stratify | **not implemented**; weak-pulse proxy is not skin tone |
| C3 multi-person | persistent one-target tracking | staged multi-face videos, identity-switch count | **not implemented** |
| C4 low resolution | VGA ≤640×480, QVGA ≤320×240 | decode genuine reduced-resolution frames before face detection | **not implemented** |
| C5 bandwidth/FPS | ≤500 kbps, ≤10 fps | re-encode actual videos at target bitrate/FPS, decode and rerun end-to-end | resamples RGB *traces* to 10 fps only |

| Test | Grounded competition target | Real experiment | Current synthetic prototype |
|---|---|---|---|
| C6 age | 60–70+ | age-stratified real data incl. older participants | **not implemented** |
| C7 occlusion | ≥40% lower face | mask actual images and verify ROI fallbacks | **not implemented** |

Do not try to “pass” the low-light task by multiplying post-extraction RGB means and claiming actual 50 lux behavior. Real cameras change shutter, gain, noise distribution and compression under weak light. Similarly, pixel downsampling after ROI averaging cannot evaluate face detector failure or changing number of pixels in the ROI.

## 4. Error metric formulas (HR in bpm)

For N paired clip predictions, `e_i = pred_i - truth_i`.

- `MAE = mean(|e_i|)`: typical absolute error.
- `RMSE = sqrt(mean(e_i²))`: outlier-sensitive error.
- `Pearson r = corr(pred, truth)`: agreement in co-variation, **not** a bias-free accuracy metric.
- `Bland--Altman bias = mean(e)`; approximate 95% limits of agreement = `bias ± 1.96*std(e, ddof=1)` when distributional assumptions permit.
- `failure_5bpm = 1(|e_i| > 5 OR invalid)`: user-defined operational measure, *not specified as the competition threshold*.
- `coverage = number_valid / number_total` and `selective MAE` for quality/abstention models; distinguish failure rate from abstention rate.

Use a separate paired bootstrap **resampled by subject, not by overlapping clip**, for 95% confidence intervals. Stratify sample sizes and report missingness. A claim of “skin-tone fairness” should also quantify the worst group and uncertainty intervals; ITA and Fitzpatrick are not equivalent concepts, and neither should be assigned merely by eyeballing a single rendered frame.

## 5. Multi-output metrics in later rounds

- **SpO2:** MAE and RMSE in **percentage points**, Bland-Altman limits, signed bias, accuracy at clinically relevant lower saturations, device/illumination generalization; do not equate narrow healthy-label prediction with reliable hypoxemia estimation.
- **BP:** separate SBP and DBP bias/MAE/RMSE in **mmHg**, cuff reference details, repeat-measure variability, calibration status, per-subject trajectories, independent subjects, and agreement plots.
- Report reference-device sampling, averaging time, recording delay, possible label synchronization issues. No substitution of hypothetical/noisy synthetic SpO2/BP labels for clinical ground truth.

## 6. Statistical comparison and literature reports

For the same held-out clips, report *paired* per-clip errors and difference distributions (e.g. POS minus GREEN). Use subject-wise bootstrap CIs; if statistical testing, pre-specify a paired test and account for multiple comparisons. Include at least one failure-case gallery with permission, raw RGB plots, frequency spectra and face-mask coverage. Avoid selecting the method by test-set tuning.

## 7. Recommended output table

Store one row per `{dataset_id, subject_id, clip_id, window_start_s, condition, method}` with:

`reference_bpm, predicted_bpm, absolute_error_bpm, failed_gt5_or_missing, quality, output_valid, roi_coverage, n_frames, actual_fps, measured_lux, bitrate_kbps, resolution, demographic_stratum, processing_ms, platform, code_version, split, reference_device, calibration_state`.

Create a `summary.csv` grouped by method × condition × subgroup, showing `N, MAE, RMSE, failure_% (defined threshold), Pearson, Bland-Altman bias/limits, coverage, time_ms` and the test protocol/version. This is compatible with judging metrics but **not the complete required competition file schema**; export official `.mat`/video/reference structures after organizers confirm details.

# 04 | Engineering roadmap and candidate research contribution

## Priorities and decision gates

| Time/order | Concrete deliverable | Gate |
|---|---|---|
| **Now, before 25 Oct 2026** | finish team formation, 3 active SPS undergraduate memberships, supervisor eligibility, signed terms, registration; email organizer for clinical dataset agreement | registration accepted + documented membership |
| Week 1 | read algorithms + download licensed/ public real HR benchmark; synchronized video/PPG loader | first valid paired subject examples |
| Week 2 | video detector + forehead/cheek ROI; GREEN/POS/CHROM in common harness | correct waveform and manual spot checks |
| Weeks 3–4 | test controlled C1/C4/C5 + build per-condition statistics and clips gallery | reproducible participant-disjoint HR table |
| Weeks 4–5 | selective / adaptive fusion prototype + ablation against fixed POS/CHROM | credible improvement on held-out subject/condition split |
| By 30 Nov 2026 | Round-1 technical IEEE-format report, code, video/CSV/.mat package per official naming, at least two constraints | pre-submission audit |
| Round 2 | clinical SpO2 labels, cross-device/low-saturation tests, documented calibration | genuine joint HR/SpO2 validation |
| Final phase | SBP + DBP labels, clinical protocol, offline/live on-device version | no overstated medical claim |

## Candidate novelty: Quality-Calibrated Multi-ROI Color Projection (QC-MRCP)

**A research hypothesis, not a validated invention or claim of novelty against all literature.**

Instead of selecting POS or CHROM globally, process several separate skin patches: forehead, left cheek, right cheek, possible periocular exposed skin if occluded. For each patch, compute candidate pulse traces `GREEN`, `POS`, `CHROM`, optionally `PBV`. Score each independently using observable *input-only* features:

- stability of dominant frequency over adjacent windows;
- amplitude/SNR of candidate HR peak relative to other frequencies;
- agreement among spatially separate ROIs (not merely agreement among algorithms sharing the same contaminated input);
- target track stability and skin-mask coverage;
- saturation, clip/dark-pixel fraction, motion magnitude and exposure changes.

Then estimate a fused pulse/HR using weights such as `w_{roi,method} = softmax(quality / temperature)` and a robust consensus rather than always trusting the peak of the loudest signal. Critically: **frequency agreement among several patches can also be caused by global flicker or whole-head motion**; independently measure the nuisance features and reject low-quality outputs. A high quality number is not a calibrated confidence probability until tested against ground truth.

### Proposed experiments

A. Fixed ROI + GREEN vs fixed ROI + POS vs fixed ROI + CHROM. B. Add adaptive ROI/patch aggregation without algorithm fusion. C. Add adaptive method fusion with weights from pre-defined non-reference features. D. Add explicit quality gate / abstention; plot risk-coverage curve and false-confidence rate. E. Re-evaluate C1, C4, C5 and C7 as *real-video* stress, holding subjects/cameras out. F. Ablate each feature, fix compute budget, and report paired bootstrap 95% CIs.

**What would support the hypothesis?** Lower worst-condition errors than fixed methods **at similar coverage**, honest improvements in false-confidence, improved subject/device generalization and sustained real-time throughput on an edge device. If it only improves a synthetic trace simulator, it remains a toy result.

## Other possible contributions (ranked)

1. **Robustness/uncertainty first:** edge-friendly POS/CHROM + multi-patch confidence and explicit abstention. Strong fit to constraints, lower compute and better interpretability.
2. **Camera-invariant color correction:** estimate nuisance exposure/white balance trends from non-skin background or reference patches *if available*. Check that this does not require unrealistic controlled calibration targets.
3. **Cross-device calibration study for SpO2:** compare fixed global RGB ratio, camera-conditioned and physiology-regularized regressors. Require broad clinical SpO2 values and confirm license/ground truth; prioritize *truthful feasibility bounds* over optimistic estimates.
4. **BP uncertainty and personalization:** if permitted by rules and user workflow, test population model against a separate **cuff-calibrated** model, with prospective recalibration and real BP-change evaluation. Do not claim absolute BP from pulse rate alone.

## Immediate fork in the road

The decision with the most value is **dataset availability**. Ask the organizers for the official clinical dataset now, while downloading a legitimate real-video HR dataset for Round 1. Do not delay the baseline until you get BP4D+; it is licensed, very large and not necessarily the exact pulse-ox ground truth needed for SpO2.

# 06 | Compliance and submission checklist

**Priority: registration deadline 25 October 2026.** Source hierarchy: 2027 FaceVitals official document for the task; uploaded September 2026 SPS Student Competition Terms & Conditions v1.4 for eligibility/ judging/travel; ask organizers when versions disagree.

## Team formation (from uploaded September 2026 Terms & Conditions)

- [ ] One eligible **faculty** supervisor (the document excludes postdocs and research associates as faculty).
- [ ] **3–10 undergraduate** team members (the document defines undergraduate as not having earned a 4-year degree).
- [ ] At least **three undergraduate SPS student members**, membership active at registration and maintained until competition end.
- [ ] At most **one graduate tutor** (optional); tutor not eligible for prizes/travel grant.
- [ ] No participant belongs to more than one team/competition simultaneously.
- [ ] Check prior SPS finalist/top-three eligibility and travel-support restrictions.
- [ ] Verify that no team members will need to change after registration deadline.
- [ ] Supervisor completes IEEE conflict-of-interest form before participation.
- [ ] Each team member reads and agrees to the terms; submit signed form per official site.
- [ ] At least **one undergraduate** can attend the ICASSP finals **in person**, subject to the organizer's published visa/exception process.
- [ ] Complete official registration by **25 October 2026**.

## Milestones (FaceVitals official challenge, accessed 19 October 2026)

| Item | Date |
|---|---|
| Registration | 25 October 2026 |
| Round 1 | 30 November 2026 |
| Round 1 results | 5 December 2026 |
| Round 2 | 16 January 2027 |
| Round 2 results | 21 January 2027 |

| Item | Date |
|---|---|
| Three finalist teams announced | 9 February 2027 |
| ICASSP 2027 | 16–21 May 2027 |

## Round-1 technical checklist

- [ ] Real RGB facial videos under controlled and uncontrolled conditions.
- [ ] Real synchronized HR reference from clinical pulse oximeter or ECG (not synthetic HR labels).
- [ ] GREEN, POS, CHROM reproducible baselines with shared preprocessing.
- [ ] Face detection, landmark ROI masks, timestamp handling and multi-video repeatability.
- [ ] **At least two constraints C1–C7** explicitly tested/addressed; explain limitations.
- [ ] MAE, RMSE, Pearson r and Bland–Altman statistics, with sample sizes and failure-case breakdown.
- [ ] Report in IEEE conference-paper format, with code and required data exports.
- [ ] Required reference videos/.mat/CSV folders and metadata according to clarified official specification.
- [ ] For any **team-collected human-subject data**, obtain institutional ethical/IRB approval and informed consent before collection, and document them as required by the challenge.

## Competition scoring (September 2026 PDF v1.4)

The uploaded terms list six equally weighted criteria, unless the official challenge document specifies otherwise: (1) innovation and thorough experiments, (2) validation-set performance, (3) test-set performance, (4) model/time complexity, (5) final report quality/clarity, (6) presentation quality/clarity. The general SPS web page currently lists a different five-criterion scheme; **obtain confirmation** of the governing rules for this competition.

## Ethics, files, and medical claims

- Do **not** collect identifiable human facial videos or vital-sign ground truth without the required consent and ethics approvals.
- Do **not** upload protected BP4D+/organizer datasets to public repositories without written permission.
- Do **not** call RGB-only SpO2 or BP estimates clinical measurements until properly validated.
- Do **not** report CSV results from the supplied synthetic trace generator as official or real-video scores.
- Retain experiment configuration, source licensing, subject-level splits, camera metadata and actual time/latency measurements.

## URLs

General SPS page: https://signalprocessingsociety.org/community-involvement/signal-processing-cup
2027 FaceVitals technical document: https://signalprocessingsociety.org/sites/default/files/2026-10/2027-SP-Cup-Official-Document-FaceVitals.pdf
SPS conflict of interest: https://www.ieee.org/about/compliance/conflict-of-interest/coiandpob.html
Organizer: vijayjeyakumar@ieee.org (copy: sp-competitions@listserv.ieee.org)